

Online Advertising with Spatial Interactions

Gagan Aggarwal

Google Research
Mountain View, California, USA
gagana@google.com

Yifan Wang*

School of Computer Science
Georgia Institute of Technology
Atlanta, Georgia, USA
ywang3782@gatech.edu

Mingfei Zhao

Google Research
Mountain View, California, USA
mingfei@google.com

Abstract

Online advertising platforms must decide how to allocate multiple ads across limited screen real estate, where each ad's effectiveness depends not only on its own placement but also on nearby ads competing for user attention. Such spatial externalities — arising from proximity, clutter, or crowding — can significantly alter welfare and revenue outcomes, yet existing auction and allocation models typically treat ad slots as independent or ordered along a single dimension.

We introduce a new framework for spatial externalities in online advertising, in which the value of an ad depends on both its slot and the configuration of surrounding ads. We model ad slots as points in a metric space, and model an advertiser's value as a function of both their bid and a discount factor determined by the configuration of other displayed ads. Within this framework, we analyze two natural models. For the Nearest-Neighbor model, where the value suppression depends only on the closest neighboring ad, we present a polynomial-time algorithm that achieves a constant approximation for the general case. We show that the allocation rule is monotone and can be implemented as a truthful mechanism. For a structured setting of 2D Euclidean space, we provide a PTAS. In contrast, for the Product-Distance model, where interference is aggregated multiplicatively across all neighbors, we establish a strong (and nearly-tight) hardness of approximation — no polynomial-time algorithm can achieve any polynomial-factor approximation unless $P=NP$, via a reduction from Max-Independent-Set.

Our results provide a foundation for reasoning about spatial externalities in ad allocation and for designing efficient, truthful mechanisms under such interactions.

CCS Concepts

• **Theory of computation** → **Approximation algorithms analysis; Algorithmic mechanism design.**

Keywords

Online Advertising, Approximation Algorithm, Mechanism Design

*This work was done while the author was visiting Google as a Student Researcher.



This work is licensed under a Creative Commons Attribution 4.0 International License.
WWW '26, Dubai, United Arab Emirates
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792370>

ACM Reference Format:

Gagan Aggarwal, Yifan Wang, and Mingfei Zhao. 2026. Online Advertising with Spatial Interactions. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792370>

1 Introduction

Online advertising, a half-trillion-dollar industry [12], represents a cornerstone of the modern digital economy. Ads are typically presented as a slate — multiple creatives shown simultaneously in distinct locations on a page — and chosen via real-time auctions. Platforms aim to maximize revenue or welfare, commonly achieved through optimizing expected yield, a metric combining the advertiser's bid with the predicted Click-Through Rate (CTR) [2, 11, 30].

A standard approach is to auction off every available ad slot as if its yield were independent of other ad slots — evaluating an ad placed in the slot independently of ads placed in other slots. In practice, webpage real estate and user attention are scarce: when multiple ads are shown together, they compete for clicks, and an ad's realized performance depends not only on its own placement but also on the number and placement of neighboring ads. The result is a web of negative externalities across the slate. This raises a natural question: *how should an ad platform allocate ads across slots when the presence of one ad can have negative externalities on the effectiveness of other ads?*

Prior work has studied several important forms of such externalities. [16] initiates the study of externalities in online advertising and introduces audience-side choice models that show that high-quality ads can steal attention from others. Cascade and Markovian scan models capture how users traverse positions and how early interactions depress engagement with lower positions [1, 23]. [20] develops a brand-effect model in position auctions, where the presence of strong brand ads generates asymmetric externalities on other advertisers. There is also a lot of empirical work that provides evidence of interference across ads. For example, [9] provides an experimental analysis of click position-bias models, showing that the performance of an ad depends critically on the surrounding slate. [19] shows how visual clutter on a webpage reduces advertising effectiveness, offering empirical evidence of inter-ad interference. While these frameworks demonstrate that externalities matter, they typically treat position effects in one dimension or at the level of the co-displayed set, abstracting away the two-dimensional layout of the page and the role of spatial proximity.

In this paper, we introduce a complementary perspective to modeling externalities in online advertising that emphasizes *spatial*

effects across ad slots. Specifically, we introduce a framework in which the value of an ad depends not only on its own slot but also on the spatial configuration of neighboring ads on the page – capturing how proximity, adjacency, and local crowding modulate ad effectiveness. This spatial-competition model departs from existing externality frameworks by making layout and distance first-class primitives. Within this framework, we design simple allocation and auction mechanisms with provable performance guarantees for both welfare and revenue, providing new insights into optimizing slates when ads can cannibalize one another through spatial crowding.

1.1 Our Model

A platform has a webpage with m ad slots. There are n advertisers competing for these ad slots, where advertiser i has a baseline valuation $v_{i,j}$ for ad slot j . We assume, without loss of generality, that $n \geq m$, since additional "virtual" advertisers with zero value ($v_{i,j} = 0$) can be introduced as needed. The platform's goal is to allocate ads to ad slots to maximize total social welfare.

If ad slots were independent, so that an advertiser's value for a slot did not depend on which other slots are filled, the problem would reduce to the standard bipartite matching problem. However, in realistic settings, ads placed in nearby positions compete for user attention, creating negative externalities across the slate. To capture this negative effect while maintaining tractability, we make the following modeling assumptions:

- The externality on an ad's realized value depends only on the set of slots selected for display, and not on which advertiser is assigned to each slot. This reflects the intuition that user attention is shaped primarily by spatial crowding rather than by the identity of advertisers shown in others slots.
- The ad slots are embedded in a metric space – for example, a two-dimensional plane representing the webpage layout – where the distance between two slots measures their spatial proximity. This captures the natural idea that the negative impact of crowding diminishes with distance, being stronger for nearby ads and weaker for distant ones.

Note that with the above assumptions, without loss we can further set up the objective function of our model as follows: let $M \subseteq [n] \times [m]$ denote a matching between advertisers and ad slots, where $(i, j) \in M$ represents that advertiser i is assigned to ad slot j . Let $S(M) = \{j | (i, j) \in M\}$ be the set of slots occupied under matching M . Our social welfare objective can be defined as

$$SW(M) := \sum_{(i,j) \in M} v_{i,j} \cdot \delta(j, S(M)), \quad (1)$$

where function $\delta(j, S(M))$ is a discount function capturing how the presence of other occupied slots $S \setminus \{j\}$ suppresses the value of placing advertiser i at slot j . Then, there must be $\delta(j, S) \in [0, 1]$, as it reflects a discount on each advertisement's value, and $\delta(j, S)$ should be a non-increasing function of S , as intuitively the more ads are placed in other slots, the greater the negative impact is on slot j . Furthermore, let $d(\cdot, \cdot) : [m] \times [m] \rightarrow [0, 1]$ be the distance function of the metric space for ad slots $[m]$, where we rescale the metric to assume that the diameter (the maximum possible distance

between any two slots) of the metric space is 1. Then, $\delta(j, S)$ should only depend on the values of $d(j, j')$ for $j' \in S \setminus \{j\}$.

We remark that our use of a *metric space* to build the objective function of the model enables a much broader modeling scope that goes beyond the physical distance between slots, as it may encode any notion of *similarity* or *substitutability* between slots. For instance, two distant slots on a webpage may serve similar user contexts, leading to high competition for user attention. By properly establishing a suitable metric embedding, we can capture the phenomenon that placing ads simultaneously in these two similar slots induces significant negative externalities. Generally speaking, as the metric can capture semantic similarity, functional overlap, or even competition for the same user intent, this flexibility makes our model a unifying abstraction that can accommodate diverse sources of inter-ad interference in online advertising.

With this flexibility in mind, the next key modeling decision is how to specify the discount function $\delta(j, S)$. Different forms of δ capture different intuitions about how crowding affects ad effectiveness. We introduce two natural instantiations that serve as canonical examples for our framework.

Nearest-Neighbor model. We define the discount factor at slot j under occupied set S by its distance to the closest other occupied slot:

$$\delta_{\min}(j, S) := \begin{cases} 1, & \text{if } S = \{j\}, \\ \min_{j' \in S \setminus \{j\}} d(j, j'), & \text{otherwise.} \end{cases}$$

This captures the intuition that the dominant interference comes from the nearest competing ad, while more distant ads have negligible effect. Since all distances $d(j, j')$ are in $[0, 1]$, it follows that $\delta_{\min}(j, S) \in [0, 1]$, and the monotonicity of $\delta_{\min}$ is guaranteed by the non-increasing property of the minimum operator. Analytically, the Nearest-Neighbor model is simpler and highlights local crowding effects.

Product-Distance model. We define the discount factor at slot j under occupied set S by the product of pairwise distances:

$$\delta_{\text{mul}}(j, S) := \begin{cases} 1, & \text{if } S = \{j\}, \\ \prod_{j' \in S \setminus \{j\}} d(j, j'), & \text{otherwise.} \end{cases}$$

Here, each additional occupied slot j' imposes an independent attenuation $d(j, j')$ on the effective value at j , and the total discount is the product of all such effects. As $d(j, j') \in [0, 1]$, we have $\delta_{\text{mul}}(j, S)$ lies in $[0, 1]$, and is monotone non-increasing in the set S . This model reflects a more global view of spatial competition, where every occupied slot contributes to the reduction in value.

Remark. We remark that the Nearest-Neighbor model and the Product-Distance model capture the two most natural extremes: one where interference is dominated by the closest neighbor, and another where every neighbor contributes multiplicatively. Other formulations are certainly possible—for example, weighted combinations or more complex attention-decay functions—but we leave

them to future work. Moreover, as we show later, even the Product-Distance model already leads to significant analytical challenges, which makes studying more general forms of δ intractable at present.

Allocation and auction. Finally, we note that our framework naturally gives rise to two distinct versions of the problem that we analyze in this paper. The first is the *allocation* version, where all valuations $v_{i,j}$ are known to the mechanism and the task is to compute a matching that maximizes the social welfare objective defined above. The second is the *auction* version, where advertisers' valuations are private inputs, and the mechanism must elicit them truthfully while ensuring that the resulting allocation is incentive compatible. Our results will address both versions, highlighting the algorithmic and mechanism-design aspects of ad allocation under spatial externalities.

1.2 Our Results

In this paper, we study the allocation and auction problems under the two spatial models introduced above: the Nearest-Neighbor model and the Product-Distance model. Our objective is to understand the best outcomes that can be achieved in polynomial time. Since computing the exact optimal allocation may require exponential time, we measure the effectiveness of algorithms by their approximation ratio with respect to the true optimum. Throughout, we present both positive and negative results for these models, highlighting the gap between what is information-theoretically possible and what can be achieved efficiently in practice.

Nearest-Neighbor model: a constant approximation. We first provide our main result for the Nearest-Neighbor model. The work most closely related to our Nearest-Neighbor model is [3], which aimed to find diverse elements (slots in our model) in metric spaces. It maps to the allocation problem in our model where all valuations $v_{i,j}$ are equal to 1. They established two key results: first, assuming the planted clique conjecture, this restricted problem admits no polynomial-time algorithm with approximation ratio better than 2; and second, they proposed an LP relaxation that achieves a $2e$ -approximation. Together, these results imply that for the Nearest-Neighbor model, constant-factor approximations are the best one can hope for.

Our first result for this model is a polynomial-time algorithm that achieves a constant approximation ratio. Moreover, the allocation rule is monotone and therefore can be converted into a truthful mechanism:

Result 1.1 (Informal Theorem 2.1 and Theorem 2.6). *For Nearest-Neighbor model, there exists a polynomial time algorithm that achieves an 18-approximation against the optimal social welfare. Furthermore, the algorithm is monotone in the bids of each advertiser and can be converted into a truthful mechanism.*

The main idea behind our approach is to generalize the LP relaxation of [3] from their simplified setting to the full Nearest-Neighbor matching model, and provide an efficient rounding algorithm for the optimal LP solution.

An additional challenge for our setting is that the rounding algorithm in [3] does not lead to a truthful auction. To design a

truthful auction, we carefully construct the probabilities in our rounding algorithm so that it can be further transformed into a truthful mechanism via Myerson's Lemma [26].

Stronger results for Nearest-Neighbor model under additional assumptions. For the most general setting of the Nearest-Neighbor model, we established a constant-factor approximation algorithm. While this resolves the question of whether constant approximations are possible, the result has two limitations.

The first limitation is about the simplicity and efficiency of the algorithm. Our algorithm is based on solving a linear programming relaxation, which makes it relatively slow in an auction environment: the LP needs to be solved polynomially many times to compute each advertiser's payment. However, in reality, when the advertisers arrive with their bids, a simpler algorithm is more appreciated to compute the matching between advertisers and slots together with the payments in a shorter time. Inspired by this, we present a simpler algorithm Nearest-Neighbor model under an extra assumption that the advertisers' valuations can be factorized:

Result 1.2 (Informal Theorem 3.1 and Theorem D.1). *For Nearest-Neighbor modTel, assuming each advertiser's valuation factorizes as $w_i \cdot u_j$, where w_i is a private parameter of advertiser i and u_j is a public value associated with slot j . Then, there exists a simple auction algorithm that achieves an $O(\log m)$ -approximation. Furthermore, if each w_i is drawn from a publicly known distribution D_i , the approximation ratio can be improved to a constant.*

The auction algorithm we run for Result 1.2 can be decomposed into two steps. Firstly, we select a subset of slots that will ultimately be used for placing ads. Since this step does not depend on advertisers' private information, the subset S can be computed offline in advance. Secondly, once the advertisers' private values w_i are revealed, we apply a simple greedy allocation: assign the advertisers in decreasing order of w_i to the slot in decreasing order of $u_j \cdot \min_{j' \in S} d(j, j')$. This allocation rule is monotone, and therefore can be converted into a truthful auction algorithm. Furthermore, the simplicity of the algorithm allows it to be efficiently implemented in practice, making it well-suited for online advertising platforms where speed is essential.

The second limitation of our $1/18$ -approximation algorithm is that the approximation factor we obtain is a bit far from 1. When the metric is captured by the Euclidean distance in the 2D plane, we provide a PTAS algorithm, which gives an approximation ratio arbitrarily close to 1:

Result 1.3 (Informal Theorem A.1). *For Nearest-Neighbor model, assuming each $v_{i,j} = 1$ and the metric $d(\cdot, \cdot)$ is given by Euclidean distances in the 2D plane, there exists a polynomial-time approximation scheme (PTAS) allocation algorithm that achieves $(1 - \epsilon)$ -approximation against the optimal allocation.*

The main idea of Result 1.3 follows from the PTAS algorithm for selecting independent disks in 2D Euclidean space [13]. Furthermore, the existence of this PTAS shows that when the metric is the Euclidean distance in 2D plane, it is strictly easier than the

general metric case – this follows from the hardness result established in [3] ruling out approximations better than 2 under general metrics.

Product-Distance model: No non-trivial approximation is achievable. In sharp contrast to the Nearest-Neighbor model, we show that in the Product-Distance model, it is hard to approximate in polynomial time. We prove the hardness by reducing the MAX-INDEPENDENT-SET (MIS) problem to the allocation problem in this model. It is well known that finding the maximum independent set on a graph $G = (V, E)$ does not admit a polynomial time algorithm that gives a $|V|^{1-\epsilon}$ approximation (e.g., see [31]). Therefore, our reduction leads to the following:

Result 1.4 (Informal Theorem B.1). *The allocation problem under the Product-Distance model admits no polynomial time approximation algorithm with ratio $m^{1-\epsilon}$ for any $\epsilon > 0$.*

Observe that if we restrict the allocation to consist of a single ad slot, then the resulting welfare is clearly an m -approximation of the optimal allocation in the Product-Distance model, as it is already an m -approximation even when the discount factor $\delta_{mul}(j, S(M))$ is not applied in the objective. Furthermore, the optimal allocation when restricting to only single slot is monotone in the bids of each advertiser, and therefore can be transformed into a truthful auction using standard Myerson’s Lemma [26]. Therefore, the hardness result together with the trivial m -approximation shows that our understanding of the Product-Distance model is essentially tight.

1.3 Related Work

Externalities in online advertising. Externalities in online advertising have been studied along several complementary dimensions. [16] initiated the study of externalities in online advertising and introduced audience-choice models to formalize how co-displayed ads compete for limited attention, with higher-quality ads siphoning clicks from others. Scan-based models capture sequential browsing behavior: in cascade and Markovian formulations, interaction with early positions depresses engagement with lower slots and breaks simple rank-by-(bid $\times$ CTR) optimality [1, 23]. Empirically, position-bias studies document that click outcomes are shaped by the surrounding slate [9], and work on visual clutter shows that crowding reduces advertising effectiveness [19] (see also recent evidence on digital clutter and memory effects by Williams and Lynch, 2024). On the mechanism-design side, externalities within keyword auctions have been analyzed both theoretically and empirically: [17] study equilibria and efficiency of GSP with externalities, while [18] combine theory with data from a large search engine to quantify their impact. Together, these strands establish that inter-ad effects matter, though most frameworks emphasize sequential position effects or set-level dependencies rather than the spatial layout and proximity effects that motivate our model.

Diverse subsets in metric space and dispersion problem. The algorithmic problem of finding the optimal allocation in our model is related to the well-studied problem of selecting diverse subsets in metric spaces and the closely related dispersion problems.

Both problems aim to select a subset of k facilities from m locations to maximize some diversity objective. [3] introduce linear-programming relaxations that yield the first constant-factor approximation for the sum-min diversity objective. [4] analyze the max-sum diversification objective, providing approximation guarantees under matroid and knapsack constraints. [7] show that local search achieves a $(1 - O(1/k))$ -approximation for max-sum diversification over negative-type metrics (and under matroid constraints). For large-scale data, [22] develop composable coresets that enable distributed or streaming approximation for both diversity and coverage objectives. Moreover, [6] design space and round-efficient MapReduce and streaming algorithms in doubling metrics. [5] further give approximation schemes specialized to the doubling-dimension setting.

In the closely related dispersion problem literature, [15] consider the objective of the average distance between selected locations. They provide a linear-time algorithm to solve the problem exactly in d -dimensional space when k is a constant, and a polynomial time approximation scheme (PTAS) when k is not fixed. [10] obtain improved constant-factor approximation algorithms for dispersion in disks. [28] derives strong upper bounds and exact algorithms for the p -dispersion-sum problem, where the goal is to maximize the distance sum between selected locations. More recently, [25] study the c -dispersion problem in general metrics, which aims to maximize the minimum distance from a point to its nearest c points. They give a simple greedy $2c$ -approximation, along with parameterized hardness results.

Unlike the aforementioned works that select a subset of locations (slots), our paper study the allocation problem between advertisers and slots. The added complexity arises because advertisers’ valuations are heterogeneous and, in mechanism design, private.

PTAS in disk graphs. Our PTAS algorithm in the nearest-neighbor model in the 2D Euclidean space is closely-related to the algorithm for selecting independent disks. In disk (and disk-like) intersection graphs, a PTAS is known to exist in classic problems such as Maximum Weight Independent Set (MWIS) and Minimum Weight Vertex Cover [14]. For unit disk graphs, [27] designed a robust PTAS for MWIS that does not require an explicit geometric embedding. [24] studied when PTASes can be upgraded to EPTAS and identified obstacles for such improvements on geometric problems, including hardness for MIS in unit disk and unit square graphs. Strengthening and systematizing these directions, [29] build a general framework for designing EPTASes for vertex-deletion problems on disk graphs, including Vertex Cover. For more general families beyond disks, [8] obtained a PTAS for MIS of pseudo-disks via local search and a constant-factor approximation for the MWIS using an LP rounding scheme.

2 Constant Approximation for Nearest-Neighbor Model

In this section, we present our algorithm for the Nearest-Neighbor model and show Result 1.1.

2.1 Constant Approximation Allocation

We first show that there exists an allocation algorithm that achieves a constant approximation for the Nearest-Neighbor model. To be specific, we prove the following theorem:

Theorem 2.1. *For Nearest-Neighbor model, there exists a polynomial time algorithm that achieves 18-approximation against the optimal social welfare.*

To begin, our first step is to reduce the problem to a geometric formulation. Formally, we define the following:

Definition 2.2 (Grouped Pseudo-Disk Selection Problem). *Let (X, d) be a metric space with 1 being an upper bound of its diameter. A pseudo-disk is a triple (c, r, w) consisting of a center $c \in X$, a radius $r \in [0, 1]$, and a weight $w \geq 0$. A pseudo-disk $D = (c, r, w)$ corresponds to the set of points in X with distance strictly smaller than r from c , and is associated with weight w .*

Given a set of pseudo-disks partitioned into K groups $T_1, T_2, \dots, T_K$. The Grouped Pseudo-Disk Selection Problem asks to find a selection T that chooses at most one pseudo-disk from each group with the objective of maximizing

$$\text{Val}(T) := \sum_{D=(c_D, r_D, w_D) \in T} w_D,$$

such that one of the following constraints is satisfied:

- *Constraint (I): Every center $c \in X$ in the metric space is covered by at most one selected disk.*
- *Constraint (II): The center of each selected disk is not covered by another selected disk.*

The Grouped Pseudo-Disk Selection problem defined in Definition 2.2 is not fully settled, as the problem may switch between Constraint (I) and Constraint (II). Clearly, Constraint (I) is stronger than Constraint (II), as every selection satisfying Constraint (I) must satisfy Constraint (II). We remark that both constraints are needed in our reduction, as we show the following:

Lemma 2.3. *The allocation problem under the Nearest-Neighbor model with n advertisers and m ad slots admits a polynomial-time reduction to a Grouped Pseudo-Disk Selection instance with $K = n$ groups of pseudo-disk and each group T_k has size $|T_k| \leq m^2$, such that the instance is a 2-approximation of the original problem. To be specific, the reduced instance satisfied the following:*

- (1) *Let M^* be the optimal matching of the Nearest-Neighbor instance and $T^* \subseteq \bigcup_{k \in [K]} T_k$ be the optimal selection of the Grouped Pseudo-Disk Selection instance that satisfies the stronger **Constraint (I)**. Then, $\text{SW}(M^*) \leq \text{Val}(T^*)$.*
- (2) *Given a selection T of the Grouped Pseudo-Disk Selection instance that satisfies the weaker **Constraint (II)**, we can find a feasible matching M for the Nearest-Neighbor instance in polynomial time, such that $\text{SW}(M) \geq 0.5 \cdot \text{Val}(T)$.*

We defer the proof of Lemma 2.3 to Section C. With the help of Lemma 2.3, to give a constant approximation, it remains to find a good selection for a Grouped Pseudo-Disk Selection instance that satisfies Constraint (II). To be specific, we show the following:

Lemma 2.4. *For a Grouped Pseudo-Disk Selection instance, let T^* be the optimal selection that satisfies **Constraint (I)**. There exists a polynomial time algorithm that finds a random selection $\tilde{T}$, such that*

- *$\tilde{T}$ satisfies **Constraint (II)**.*
- $\mathbb{E}[\text{Val}(\tilde{T})] \geq \frac{1}{9} \cdot \text{Val}(T^*)$.

We defer the proof of Lemma 2.4 to Section 2.2, and first prove Theorem 2.1 via combining Lemma 2.3 and Lemma 2.4:

PROOF OF THEOREM 2.1. Given a Nearest-Neighbor instance, we first apply Lemma 2.3 to reduce the problem to a Grouped Pseudo-Disk Selection instance. Then, we run the algorithm provided by Lemma 2.4 to find $\tilde{T}$, and use Property (2) of Lemma 2.3 to convert it back to a feasible solution $\tilde{M}$. By Lemmas 2.3 and 2.4, we have

$$\mathbb{E}[\text{SW}(\tilde{M})] \geq 0.5 \cdot \mathbb{E}[\text{SW}(\tilde{T})] \geq \frac{1}{18} \cdot \text{Val}(T^*) \geq \frac{1}{18} \cdot \text{SW}(M^*),$$

where M^* is the optimal solution for the Nearest-Neighbor instance, and T^* is the optimal selection of the Grouped Pseudo-Disk Selection instance that satisfies Constraint (I). $\square$

2.2 Proof of Lemma 2.4 via LP Relaxation

In this subsection, we provide the proof of Lemma 2.4. Our main idea is to relax the Grouped Pseudo-Disk Selection problem with Constraint (I) to an LP, and further round the LP to give a solution that satisfies Constraint (II).

Let $D_1, \dots, D_N$ be the pseudo disks, where $D_i = (c_i, r_i, w_i)$. Let $T_1, \dots, T_k$ be the group of disks, such that each $i \in [N]$ belongs to exactly one T_k . Consider the following LP:

$$\begin{aligned} & \text{maximize} && \sum_{i \in [N]} w_i \cdot x_i, \\ & \text{s.t.} && \forall c \in X \quad \sum_{i \in [N]: d(v, c_i) < r_i} x_i \leq 1 && (\text{LP}_{\text{RELAX}}) \\ & && \forall k \in [K] \quad \sum_{i \in [N]: i \in T_k} x_i \leq 1 \\ & && \forall i \in [N] \quad x_i \in [0, 1] \end{aligned}$$

Lemma 2.5. *LP_{RELAX} is a feasible relaxation of the Grouped Pseudo-Disk Selection problem with Constraint (I), that is, given a Grouped Pseudo-Disk Selection instance, the objective of LP_{RELAX} is at least $\text{Val}(T^*)$, where T^* is the optimal selection that satisfies Constraint (I).*

PROOF. It's sufficient to show that solution $\{\tilde{x}_i\}_{i \in [N]}$ with $\tilde{x}_i = 1[i \in T^*]$ is a feasible integral solution of LP_{RELAX} . For the first constraint, since T^* is feasible with Constraint (I), for every $c \in X$, $\sum_{i: d(v, c_i) < r_i} \tilde{x}_i \leq 1$ holds. For the second constraint, since T^* selects at most one disk from each group T_k , $\sum_{i \in T_k} \tilde{x}_i \leq 1$ is satisfied. Therefore, $\tilde{x}_i = 1[i \in T^*]$ is a feasible integral solution of LP_{RELAX} . $\square$

With LP_{RELAX} , we now present our main Algorithm 1, which is the desired algorithm for Lemma 2.4.

Algorithm 1: CONSTANT APPROXIMATION FOR GROUP PSEUDO-DISK SELECTION

Input: Disks $D_1, \dots, D_N$ and groups $T_1, \dots, T_k$
 Solve LP_{RELAX} . Let x_i^* be the optimal solution.
 Initiate $T = \emptyset$ and $\tilde{T} = \emptyset$.
 For each $i \in N$, add i into T with probability $x_i^*/3$.
for $i \in T$ **do**
 Let $k \in [K]$ be the group that satisfies $i \in T_k$.
 if $\nexists i' \in T : i' \neq i \wedge (i' \in T_k \vee d(c_i, c_{i'}) < r_{i'})$ **then**
 Add i into $\tilde{T}$ with probability $\frac{1}{3} \cdot \frac{1}{\prod_{i' \neq i} (1 - 1[i' \in T_k \vee d(c_i, c_{i'}) < r_{i'}] \cdot x_{i'}^*/3)}$ (2)
end
end
Output: Feasible selection $\tilde{T}$

To show Algorithm 1 is the desired algorithm, we first show that Algorithm 1 is a feasible algorithm, and then show that the expectation of $\text{Val}(\tilde{T})$ is at least $\frac{1}{9} \cdot \text{Val}(T^*)$.

Feasibility of Algorithm 1. To show Algorithm 1 is feasible, we need to show

- The probability defined in (2) is a feasible probability that falls in $[0, 1]$.
- The output $\tilde{T}$ satisfies Constraint (II).

For the feasibility of (2), clearly the probability in (2) must be non-negative. On the other hand, note that

$$\begin{aligned} & \prod_{i' \neq i} (1 - 1[i' \in T_k \vee d(c_i, c_{i'}) < r_{i'}] \cdot x_{i'}^*/3) \\ & \geq 1 - \sum_{i' \in T_k \setminus \{i\}} \frac{x_{i'}^*}{3} - \sum_{i' \neq i: d(c_i, c_{i'}) < r_{i'}} \frac{x_{i'}^*}{3} \\ & \geq 1 - \frac{1}{3} - \frac{1}{3} \geq \frac{1}{3}, \end{aligned}$$

where the first inequality follows from union bound, and the second inequality follows from the constraints in $(\text{LP}_{\text{RELAX}})$. Therefore, the probability in (2) is upper-bounded by 1.

For the feasibility of $\tilde{T}$, note that a disk i is possible to be added into $\tilde{T}$ only when it's in T , and its center is not covered by another disk in T . Therefore, the center of every disk in $\tilde{T}$ is not covered by a second disk, and therefore Constraint (II) is satisfied.

The value of $\tilde{T}$. Now, we show $\mathbb{E}[\text{Val}(\tilde{T})] \geq \frac{1}{9} \cdot \text{Val}(T^*)$. To achieve this, we show a stronger claim that

$$\mathbb{E}[\text{Val}(\tilde{T})] = \frac{1}{9} \cdot \text{objective of } (\text{LP}_{\text{RELAX}}) \geq \frac{1}{9} \cdot \text{Val}(T^*), \quad (3)$$

where the inequality is guaranteed by Lemma 2.5. To show the equality in (3), note that for each disk i , we have

$$\begin{aligned} \Pr[i \in \tilde{T}] &= \Pr[i \in T] \cdot \Pr[\nexists i' \neq i \in T : i' \in T_k \vee d(c_i, c_{i'}) < r_{i'}] \cdot (2) \\ &= \frac{x_i^*}{3} \cdot \frac{1}{3} \cdot \frac{\prod_{i' \neq i} (1 - 1[i' \in T_k \vee d(c_i, c_{i'}) < r_{i'}] \cdot x_{i'}^*/3)}{\prod_{i' \neq i} (1 - 1[i' \in T_k \vee d(c_i, c_{i'}) < r_{i'}] \cdot x_{i'}^*/3)} \\ &= \frac{x_i^*}{9}, \end{aligned}$$

where the first equality follows from the fact that whether $i \in T$, whether i satisfies the constraint in Line 6 of Algorithm 1, and whether i is added into $\tilde{T}$ in Line 7 of Algorithm 1 are independent events. By multiplying w_i to both sides of the above equality and summing over all disks $i \in [N]$, we have $\mathbb{E}[\text{Val}(\tilde{T})]$ equals to $\frac{1}{9}$ times the objective of $(\text{LP}_{\text{RELAX}})$.

Efficiency of Algorithm 1. Finally, we need to show Algorithm 1 is efficient. Since there are $O(|X| + K + N)$ constraints and $O(N)$ variables in $(\text{LP}_{\text{RELAX}})$, it requires $O(\text{poly}(|X|, K, N))$ time to solve $(\text{LP}_{\text{RELAX}})$. For the remaining steps in Algorithm 1, it can be clearly verified that these steps run in $O(\text{poly}(|X|, K, N))$ time. Therefore, Algorithm 1 is efficient.

2.3 Converting the Algorithm to a Truthful Mechanism

Now, we show how to convert the allocation we discussed above to a truthful mechanism. More specifically, we prove that the allocation algorithm in Theorem 2.1 is monotone, so that the Myerson's Lemma can be applied. We defer the proof of Theorem 2.6 to Section C.

Theorem 2.6. *The allocation algorithm for Nearest-Neighbor model for Theorem 2.1 can be converted into a truthful mechanism.*

3 Simple Algorithms for Nearest-Neighbor Model

In this section, we present efficient and simple algorithms for the special case of the Nearest-Neighbor model, assuming value $v_{i,j}$ can be factorized as $v_{i,j} = w_i \cdot u_j$, where u_j is publicly known. Similar to Theorem 2.6, we also show that the algorithms can be converted into truthful mechanisms.

The main intuition for algorithms in this section is *simultaneous optimization*. The concept of simultaneous optimization is defined for a combinatorial optimization problem, where the objective is to optimize an unknown norm. The goal of simultaneous optimization is to provide a feasible solution for the combinatorial optimization problem, such that the solution is approximately optimal for every norm in a certain class. We show that this *factorized* Nearest-Neighbor model with the assumption that $v_{i,j} = w_i \cdot u_j$ can be modeled as a simultaneous optimization problem over a class of norms parameterized by the unknown advertisers' values $\{w_i\}$. When the values $\{w_i\}$ are arbitrary, we present a direct proof that gives an $O(\log m)$ approximation. When the value $\{w_i\}$ are stochastic such that $w_i \sim \mathcal{D}_i$, we show that it's sufficient to use the expectations of each distribution as parameters that define the norm of the problem, and the optimal selection of slots under this specific norm achieves a constant approximation.

Due to space limit, in this section, we focus on the case when advertisers' values are adversarial, and defer the stochastic advertisers' values case to Section D. For adversarial advertisers' values, we prove the following:

Theorem 3.1. *For Nearest-Neighbor model, assuming each advertiser's valuation factorizes as $w_i \cdot u_j$, where w_i is a private value of advertiser i and u_j is a public weight associated with slot j . Then,*

there exists an allocation algorithm running in $\tilde{O}(n + m)$ time that achieves an $O(\log m)$ -approximation. Furthermore, the algorithm can be converted to a truthful mechanism.

To prove Theorem 3.1, we show Algorithm 2 is the desired algorithm for Theorem 3.1. Throughout the subsection, we assume the number of slots m is at most the number of advertisers n , i.e., $m \leq n$. This is without loss of generality, as we may arbitrarily add dummy advertisers with value 0 into the instance.

Algorithm 2: $O(\log m)$ APPROX. FOR FACTORIZED NEAREST-NEIGHBOR MODEL

Input: Number of slots m and advertisers n , distance function d

/* Slot Selection Phase */

Input slot weights $u_1, u_2, \dots, u_m$, such that WLOG $u_1 \geq u_2 \geq \dots \geq u_m$.

Let $L = \lceil \log_2(m^2) \rceil$.

Choose $r \in \{2^{-L}, 2^{-L+1}, \dots, 2^{-1}\}$ uniformly at random.

Initiate slot set $\tilde{S} = \emptyset$.

for $j = 1 \rightarrow m$ **do**

if $d(j, j') \geq r$ holds for every $j' \in \tilde{S}$ **then**

Add j into $\tilde{S}$

end

end

/* Allocation Phase */

Advertisers report values $w_1, w_2, \dots, w_n$.

For $i \in [\tilde{S}]$, assign the advertiser with i -th largest value to the slot in $\tilde{S}$ such that the value of $u_j \cdot \min_{j' \in \tilde{S} \setminus \{j\}} d(j, j')$ is the i -largest.

Let $\tilde{M}$ be the corresponding matching.

Output: Matching $\tilde{M}$.

Representing the objective as a norm. To show that Algorithm 2 achieves an $O(\log m)$ approximation, we first introduce some extra notations that will be used in our proof.

Given an instance of the factorized Nearest-Neighbor model where $w_1, w_2, \dots, w_n$ are values of advertisers and $u_1 \geq u_2 \geq \dots \geq u_m \geq 0$ are weights of slots. We define

$$W := (w_1, \dots, w_n)$$

to be the value vector of advertisers.

For an m -dimensional vector $Z = (z_1, z_2, \dots, z_m) \in \mathbb{R}_{\geq 0}^m$ that only contains non-negative values, we define notation

$$Z^\downarrow := (z_1^\downarrow, \dots, z_m^\downarrow)$$

to be a vector denoting the sequence obtained by ordering values in Z in non-increasing order, where notation $z_i^\downarrow$ represents the i -th largest value in Z . Then, for $Z \in \mathbb{R}_{\geq 0}^m$, we define function $f : \mathbb{R}_{\geq 0}^m \rightarrow \mathbb{R}_{\geq 0}$ parametrized by $W^\downarrow$ as follows (recall that $n \geq m$):

$$f(Z; W^\downarrow) := \sum_{j=1}^m z_j^\downarrow \cdot w_j^\downarrow.$$

Then, the following claim shows that the function f is a norm:

Claim 3.2. Function $f(Z, W^\downarrow)$ is a monotone symmetric norm.

We defer the proof of Claim 3.2 to Section D. Now, we show how to represent the objectives of the optimal matching and the matching obtained by Algorithm 2 as a function of f . For any subset $S \subseteq [m]$ of slots, we define vectors $U(S)$ and $Z(S)$, such that

$$(U(S))_j := u_j \cdot \mathbf{1}[j \in S] \text{ and } (Z(S))_j := \mathbf{1}[j \in S] \cdot u_j \cdot \min_{j' \in S \setminus \{j\}} d(j, j'),$$

where notation $\mathbf{1}[\cdot]$ represents the indicator function, and similarly $U(S), Z(S)$ to be vectors denoting the sequence obtained by ordering values in $U(S)$ and $Z(S)$ in non-increasing order, respectively.

Let M^* be the optimal matching of the factorized Nearest-Neighbor instance. With the above notations, we can simplify the value of $SW(M^*)$ as

$$SW(M^*) = \max_{S \subseteq [m]} f(Z(S); W^\downarrow). \quad (4)$$

On the other hand, For $l \in \{0, 1, 2, \dots, L = \lceil \log_2(m^2) \rceil\}$, we let $\tilde{S}_l$ be the set of slots selected in the slot selection phase of Algorithm 2 and $\tilde{M}_l$ be the corresponding matching, assuming the parameter r from Line 3 of Algorithm 2 equals to 2^{-l} . Then, we have

$$SW(\tilde{M}_l) = f(Z(\tilde{S}_l); W^\downarrow). \quad (5)$$

Proof of the approximation ratio. Since the parameter r in Algorithm 2 is chosen uniformly from $\{0, 1, \dots, L\}$, the expected performance of Algorithm 2 is exactly $\sum_{l=0}^L SW(\tilde{M}_l)$. Since $L = \lceil \log_2(m^2) \rceil = O(\log m)$, to show that Algorithm 2 achieves an $O(\log m)$ approximation, it's sufficient to show

$$\mathbb{E}[SW(\tilde{M})] = \frac{1}{L+1} \cdot \sum_{l=0}^L SW(\tilde{M}_l) \geq \frac{1}{L+1} \cdot \frac{1}{4} \cdot SW(M^*).$$

By applying (4) and (5) to the above inequality, it's equivalent to show that for any $S \subseteq [m]$, we have

$$4 \cdot \sum_{l=0}^L f(Z(\tilde{S}_l); W^\downarrow) \geq f(Z(S); W^\downarrow). \quad (6)$$

Fix the subset S on the RHS of (6). We first partition S into

$$S = \left(\bigcup_{l=0}^{L-1} S_l \right) \cup S_L,$$

such that for $l \in \{1, 2, \dots, L-1\}$, we have

$$S_l := \left\{ j \in S : \min_{j' \in S \setminus \{j\}} d(j, j') \in (2^{-l}, 2^{-l+1}] \right\},$$

and $S_L := S \setminus \bigcup_{l=1}^{L-1} S_l$ includes all remaining items j that satisfies $\min_{j' \in S \setminus \{j\}} d(j, j') \leq 2^{-L+1}$.

For $j \in S_l$, as the value of $\min_{j' \in S \setminus \{j\}} d(j, j')$ is upper-bounded by 2^{-l+1} , its contribution to $SW(M^*)$ can be bounded by $u_j \cdot 2^{-l+1}$ times the value of the advertiser matched to j . Applying this inequality to the function $f(Z(S); W^\downarrow)$, we have

$$f(Z(S); W^\downarrow) \leq \sum_{l=1}^L f(U(S_l); W^\downarrow) \cdot 2^{-l+1}. \quad (7)$$

Similarly, for $\tilde{S}_l$ obtained by Algorithm 2, note that Algorithm 2 adds j into $\tilde{S}_l$ only when j is at least $r = 2^{-l}$ away from existing slots

in $\tilde{S}_l$. Therefore, for $j \in \tilde{S}_l$, there must be $\min_{j' \in \tilde{S}_l \setminus \{j\}} d(j, j') \geq 2^{-l}$, which further implies

$$f(Z(\tilde{S}_l); W^\downarrow) \geq f(U(\tilde{S}_l); W^\downarrow) \cdot 2^{-l} \quad (8)$$

Then, the following claim, which serves as the key idea to prove (6), relates the value of $f(U(S_l); W^\downarrow)$ to the value of $f(U(\tilde{S}_{l+1}); W^\downarrow)$:

Claim 3.3. For $l \in \{1, 2, \dots, L-1\}$, we have

$$f(U(S_l); W^\downarrow) \leq 2 \cdot f(U(\tilde{S}_{l+1}); W^\downarrow).$$

PROOF. For subset S_l , we partition it into $S_l^{(1)} \cup S_l^{(2)}$, such that $S_l^{(1)} = S_l \cap \tilde{S}_{l+1}$, and $S_l^{(2)} = S_l \setminus \tilde{S}_{l+1}$. We further define vector $U_1 = U(S_l^{(1)})$, and $U_2 = U(S_l^{(2)})$.

Since $S_l^{(1)} \subseteq U(\tilde{S}_{l+1})$, the vector U_1 is dominated by $U(\tilde{S}_{l+1})$. By monotonicity of function f , this implies

$$f(U_1; W^\downarrow) \leq f(U(\tilde{S}_{l+1}); W^\downarrow).$$

Next, we show $f(U_2; W^\downarrow)$ is also upper-bounded by $f(U(\tilde{S}_{l+1}); W^\downarrow)$. We achieve this by building up a mapping from $S_l^{(2)}$ to $\tilde{S}_{l+1}$. For $j \in S_l^{(2)}$, since we have $j \notin \tilde{S}_{l+1}$, there must exist $j' \in \tilde{S}_{l+1}$ that satisfies $d(j, j') < r = 2^{-l-1}$. Furthermore, j' should satisfy $u_{j'} \geq u_j$, as Algorithm 2 considers each slot in decreasing order of u_j . In this case, we call $j \in S_l^{(2)}$ is “blocked” by $j' \in \tilde{S}_{l+1}$.

The key observation of our proof is that for $j_1, j_2 \in S_l^{(2)}$ such that $j_1 \neq j_2$, let $j'_1, j'_2 \in \tilde{S}_{l+1}$ be the slots that block j_1 and j_2 respectively. Then, there must be $j'_1 \neq j'_2$. We prove via contradiction: assuming $j'_1 = j'_2$, which means $d(j_1, j'_1) < 2^{-l-1}$, and $d(j_2, j'_1) < 2^{-l-1}$. By triangle inequality, there must be $d(j_1, j_2) \leq 2 \cdot 2^{-l-1} = 2^{-l}$, which is in contrast to the assumption that $j_1, j_2 \in S_l$, as it implies $d(j_1, j_2) > 2^{-l}$.

The above observation implies that there exists a mapping from non-zero values in U_2 to non-zero values in $U(\tilde{S}_{l+1})$, such that each non-zero value in U_2 is mapped to a value that is no smaller than itself, and each value in $U(\tilde{S}_{l+1})$ is mapped at most once. By rearranging the values in U_2 (which is allowed as f is symmetric) and monotonicity of function f , we have

$$f(U_2; W^\downarrow) \leq f(U(\tilde{S}_{l+1}); W^\downarrow).$$

Combining the above two inequalities together gives

$$f(U(S_l); W^\downarrow) \leq f(U_1; W^\downarrow) + f(U_2; W^\downarrow) \leq 2 \cdot f(U(\tilde{S}_{l+1}); W^\downarrow),$$

where the first inequality follows from the triangle inequality of function f . $\square$

Combining (7), (8), and Claim 3.3, we get

$$\begin{aligned} f(Z(S); W^\downarrow) &\leq \sum_{l=1}^L f(U(S_l); W^\downarrow) \cdot 2^{-l+1} \\ &\leq f(U(S_L); W^\downarrow) \cdot 2^{-L+1} + 2 \cdot \sum_{l=2}^L f(U(\tilde{S}_l); W^\downarrow) \cdot 2^{-l+1} \\ &\leq f(U(S_L); W^\downarrow) \cdot 2^{-L+1} + 4 \cdot \sum_{l=2}^L f(Z(\tilde{S}_l); W^\downarrow) \end{aligned}$$

$$\begin{aligned} &\leq 4 \cdot \sum_{l=0}^L f(Z(\tilde{S}_l); W^\downarrow) \\ &\quad + \left(f(U(S_L); W^\downarrow) \cdot 2^{-L+1} - 4 \cdot f(Z(\tilde{S}_0); W^\downarrow) \right). \end{aligned}$$

Therefore, to show (6), it only remains to show

$$f(U(S_L); W^\downarrow) \cdot 2^{-L+1} \leq 4 \cdot f(Z(\tilde{S}_0); W^\downarrow). \quad (9)$$

By (8), we have

$$f(Z(\tilde{S}_0); W^\downarrow) \geq f(U(\tilde{S}_0); W^\downarrow) \cdot 1 \geq u_1 \cdot w_1,$$

where we recall u_1 represents the highest slot weight, and w_1 represents the highest advertiser's value. On the other hand, we have

$$f(U(S_L); W^\downarrow) \cdot 2^{-L+1} \leq 2 \cdot \frac{1}{m^2} \cdot m \cdot (u_1 \cdot w_1) \leq 2u_1 \cdot w_1,$$

where the first inequality uses the fact that each value in vector $U(S_L)$ is upper-bounded by u_1 , and similarly each value in vector $W^\downarrow$ is upper-bounded by w_1 . Combining the above two inequalities together shows (9).

Truthfulness and Efficiency. Finally, we finish the proof of Theorem 3.1 by showing that Algorithm 2 is efficient, and can be converted into a truthful mechanism via Myerson's Lemma [26]. By Algorithm 2, the allocation phase only requires to sort the advertisers and the slots respectively, and match the advertisers and slots in order. Therefore, the algorithm runs in $\tilde{O}(n + m)$ time.

For the truthfulness of the algorithm, we only need to show that the objective achieved in the allocation phase is monotone. Fix $\tilde{S}$ to be the set of pre-selected slots, and let $W = (w_1, w_2, \dots, w_n)$ be the vector of advertisers' values. The objective achieved in the allocation phase is exactly

$$\sum_{i=1}^m (Z^\downarrow(\tilde{S}))_i \cdot w_i^\downarrow,$$

where $w_i^\downarrow$ represents the i -th largest value in vector W . Now suppose we increase the value of w_i to w'_i , while leave the remaining values unchanged. Let W' be the modified value vector, and $w_i'^\downarrow$ be the i -th largest value in vector W' . A crucial observation is that there must be $w_i'^\downarrow \geq w_i^\downarrow$ for every $i \in [n]$. Then, we have

$$\sum_{i=1}^m (Z^\downarrow(\tilde{S}))_i \cdot w_i'^\downarrow \geq \sum_{i=1}^m (Z^\downarrow(\tilde{S}))_i \cdot w_i^\downarrow,$$

i.e., the objective achieved by vector W' is at least the objective achieved by vector W . Therefore, the allocation phase of Algorithm 2 is monotone, which implies the algorithm can be converted into a truthful auction algorithm via Myerson's Lemma.

References

- [1] Gagan Aggarwal, Jon Feldman, S. Muthukrishnan, and Martin Pál. 2008. Sponsored Search Auctions with Markovian Users. In *Internet and Network Economics (WINE) (Lecture Notes in Computer Science, Vol. 5385)*. Springer, 621–628. doi:10.1007/978-3-540-92185-1_68
- [2] Gagan Aggarwal, Ashish Goel, and Rajeev Motwani. 2006. Truthful Auctions for Pricing Search Keywords. In *Proceedings of the 7th ACM Conference on Electronic Commerce (EC '06)* (Ann Arbor, Michigan, USA), Joan Feigenbaum, John C.-I. Chuang, and David M. Pennock (Eds.). Association for Computing Machinery, 1–7. doi:10.1145/1134707.1134708
- [3] Aditya Bhaskara, Mehrdad Ghadiri, Vahab S. Mirrokni, and Ola Svensson. 2016. Linear Relaxations for Finding Diverse Elements in Metric Spaces. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5–10, 2016, Barcelona, Spain*, Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (Eds.). 4098–4106. <https://proceedings.neurips.cc/paper/2016/hash/d79c6256b9bdac53a55801a066b70da3-Abstract.html>
- [4] Allan Borodin, Aadhar Jain, Hyun Chul Lee, and Yuli Ye. 2017. Max-Sum Diversification, Monotone Submodular Functions and Dynamic Updates. *ACM Transactions on Algorithms* 13, 3 (2017), 41:1–41:25. doi:10.1145/3086464
- [5] Matteo Ceccarello, Andrea Pietracaprina, and Geppino Pucci. 2018. Diversity Maximization in Doubling Metrics. In *29th International Symposium on Algorithms and Computation (ISAAC 2018) (LIPIcs, Vol. 123)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 38:1–38:13. doi:10.4230/LIPIcs.ISAAC.2018.38
- [6] Matteo Ceccarello, Andrea Pietracaprina, Geppino Pucci, and Eli Upfal. 2017. MapReduce and Streaming Algorithms for Diversity Maximization in Metric Spaces of Bounded Doubling Dimension. *Proceedings of the VLDB Endowment* 10, 5 (2017), 469–480. <https://www.vldb.org/pvldb/vol10/p469-ceccarello.pdf>
- [7] Alfonso Cevallos, Friedrich Eisenbrand, and Rico Zenklusen. 2019. An Improved Analysis of Local Search for Max-Sum Diversification. *Mathematics of Operations Research* 44, 4 (2019), 1494–1509. doi:10.1287/moor.2018.0965
- [8] Timothy M. Chan and Sarel Har-Peled. 2012. Approximation Algorithms for Maximum Independent Set of Pseudo-Disks. *Discrete & Computational Geometry* 48, 2 (2012), 373–392. doi:10.1007/s00454-012-9417-5
- [9] Nick Craswell, Onno Zoeter, Michael Taylor, and Bill Ramsey. 2008. An Experimental Comparison of Click Position-Bias Models. In *Proceedings of the 2008 International Conference on Web Search and Data Mining (WSDM)*. ACM, 87–94. doi:10.1145/1341531.1341545
- [10] Adrian Dumitrescu and Minghui Jiang. 2012. Dispersion in disks. *Theory of Computing Systems* 51, 2 (2012), 128–142.
- [11] Benjamin Edelman, Michael Ostrovsky, and Michael Schwarz. 2007. Internet Advertising and the Generalized Second-Price Auction. *American Economic Review* 97, 1 (2007), 242–259. <https://www.aeaweb.org/articles?id=10.1257/aer.97.1.242>
- [12] Insider Intelligence eMarketer. 2023. Digital ad spend worldwide will pass \$600 billion this year. <https://www.emarketer.com/content/digital-ad-spend-worldwide-pass-600-billion-this-year> Accessed 2025-10-05.
- [13] Thomas Erlebach, Klaus Jansen, and Eike Seidel. 2005. Polynomial-Time Approximation Schemes for Geometric Intersection Graphs. *SIAM J. Comput.* 34, 6 (2005), 1302–1323. doi:10.1137/S0097539702402676
- [14] Thomas Erlebach, Klaus Jansen, and Eike Seidel. 2005. Polynomial-Time Approximation Schemes for Geometric Intersection Graphs. *SIAM J. Comput.* 34, 6 (2005), 1302–1323. doi:10.1137/S0097539702402676
- [15] Sándor P. Fekete and Henk Meijer. 2004. Maximum Dispersion and Geometric Maximum Weight Cliques. *Algorithmica* 38, 3 (2004), 501–511. doi:10.1007/s00453-003-1060-4
- [16] Arpita Ghosh and Mohammad Mahdian. 2008. Externalities in Online Advertising. In *Proceedings of the 17th International World Wide Web Conference (WWW)*. ACM, 161–168. doi:10.1145/1367497.1367520
- [17] Ioannis Giotis and Anna R. Karlin. 2008. On the Equilibria and Efficiency of the GSP Mechanism in Keyword Auctions with Externalities. In *Internet and Network Economics (WINE 2008)*, LNCS 5385. 629–638. https://link.springer.com/chapter/10.1007/978-3-540-92185-1_69
- [18] Renato Gomes, Nicole Immorlica, and Evangelos Markakis. 2009. Externalities in Keyword Auctions: An Empirical and Theoretical Assessment. In *Proceedings of the 5th International Workshop on Internet and Network Economics (WINE 2009) (Lecture Notes in Computer Science, Vol. 5929)*, Stefano Leonardi (Ed.). Springer, Berlin, Heidelberg, 172–183. doi:10.1007/978-3-642-10841-9_17
- [19] Peter Hammer, Erica Riebe, and Rachel Kennedy. 2009. How Clutter Affects Advertising Effectiveness. *Journal of Advertising Research* 49, 2 (2009), 159–163. doi:10.2501/S0021849909090217
- [20] Patrick Hummel and R. Preston McAfee. 2014. Position Auctions with Externalities. In *Web and Internet Economics (WINE) (Lecture Notes in Computer Science, Vol. 8877)*. Springer, 417–422. doi:10.1007/978-3-319-13129-0_33
- [21] Sharat Ibrahimpur and Chaitanya Swamy. 2020. Approximation Algorithms for Stochastic Minimum-Norm Combinatorial Optimization. In *61st IEEE Annual Symposium on Foundations of Computer Science, FOCS 2020, Durham, NC, USA, November 16–19, 2020*, Sandy Irani (Ed.). IEEE, 966–977. doi:10.1109/FOCS46700.2020.00094
- [22] Piotr Indyk, Sepideh Mahabadi, Mohammad Mahdian, and Vahab S. Mirrokni. 2014. Composable core-sets for diversity and coverage maximization. In *PODS 2014*. ACM, 100–108. doi:10.1145/2594538.2594560
- [23] David Kempe and Mohammad Mahdian. 2008. A Cascade Model for Externalities in Sponsored Search. In *Internet and Network Economics (WINE) (Lecture Notes in Computer Science, Vol. 5385)*. Springer, 585–596. doi:10.1007/978-3-540-92185-1_65
- [24] Dániel Marx. 2005. Efficient Approximation Schemes for Geometric Problems?. In *European Symposium on Algorithms (ESA 2005) (Lecture Notes in Computer Science, Vol. 3669)*. Springer, 448–459.
- [25] Pawan K. Mishra and Gautam K. Das. 2021. Approximation Algorithms For The Dispersion Problems in a Metric Space. *arXiv preprint arXiv:2105.09313* (2021). doi:10.48550/arXiv.2105.09313
- [26] Roger B. Myerson. 1981. Optimal Auction Design. *Mathematics of Operations Research* 6, 1 (1981), 58–73. doi:10.1287/moor.6.1.58
- [27] Tim Nieberg, Johann L. Hurink, and Walter Kern. 2004. A Robust PTAS for Maximum Weight Independent Sets in Unit Disk Graphs. In *Graph-Theoretic Concepts in Computer Science (WG 2004) (Lecture Notes in Computer Science, Vol. 3353)*. Springer, 214–221. doi:10.1007/978-3-540-30559-0_18
- [28] David Pisinger. 2006. Upper Bounds and Exact Algorithms for p-Dispersion Problems. *Computers & Operations Research* 33, 5 (2006), 1380–1398. doi:10.1016/j.cor.2004.09.017
- [29] Erik Jan van Leeuwen. 2006. Better Approximation Schemes for Disk Graphs. In *Algorithm Theory – SWAT 2006 (Lecture Notes in Computer Science, Vol. 4059)*. Springer, 316–327. doi:10.1007/11785293_30
- [30] Hal R. Varian. 2007. Position Auctions. *International Journal of Industrial Organization* 25, 6 (2007), 1163–1178. <https://people.ischool.berkeley.edu/~hal/Papers/2006/position.pdf>
- [31] David Zuckerman. 2007. Linear Degree Extractors and the Inapproximability of Max Clique and Chromatic Number. *Theory of Computing* 3, 6 (2007), 103–128. doi:10.4086/toc.2007.v003a006

A PTAS Algorithm for Unweighted Advertisers

In this section, we show that there exists a PTAS algorithm that achieves a $(1+\epsilon)$ -approximation for the factorized Nearest-Neighbor model with 2D Euclidean metric and unweighted advertisers, that is, each valuation $v_{i,j}$ equals to u_j , which only depends on the identity of a slot, and the metric space is the Euclidean distance in the 2D plane. To be specific, we show the following:

Theorem A.1. *For Nearest-Neighbor model with n advertisers and m slots, assuming each $v_{i,j} = u_j$ and the metric $d(\cdot, \cdot)$ is given by Euclidean distances in the 2D plane, given $\epsilon \in (0, 0.5)$, there exists an algorithm that runs in $O(m^{O(\epsilon^{-2})} \cdot \text{poly}(n, \epsilon^{-1}))$ time and achieves a $(1 + \epsilon)$ -approximation against the optimal allocation.*

Proof sketch. We defer the detailed proof of Theorem A.1 to the full version of the paper due to space limit, and only provide a proof sketch. Our algorithm consists of two steps. First, we reduce the problem to a Weighted Disk Selection problem. This problem is similar to the Grouped Pseudo-Disk Selection problem defined in Definition 2.2 with Constraint (II), where the metric is the Euclidean distance on a 2D plane. The main difference is that, in the Weighted Disk Selection problem, the disks are not partitioned into multiple groups that restrict the selection to at most one disk per group. Instead, we only require that the total number of selected disks be bounded. Next, we solve the grouped pseudo-disk selection problem via a PTAS algorithm. Our main algorithm is a dynamic programming algorithm similar to the algorithm in [13] for selecting independent disks on a 2D plane. We first group disks by size into logarithmically many levels and overlay shifted grids on the plane. By carefully choosing a grid shift, we can remove only a small fraction of disks—losing at most a $(1 - \epsilon)$ factor of

the optimal value. The grids at different levels form a hierarchical decomposition of the plane, where each cell interacts only with a constant-sized “interface” of neighboring disks. This property allows us to design a dynamic-programming algorithm that processes the grid hierarchy bottom-up, combining locally feasible disk selections while ensuring global feasibility. Since the number of relevant configurations per cell is polynomial for any fixed ϵ , the algorithm runs in $N^{O(\text{poly}(\epsilon^{-1}))}$ time and returns a solution whose total weight is within a $(1 - \epsilon)$ factor of the optimum.

B Hardness of Product-Distance Model

In this section, we prove the hardness result for the Product-Distance model. To be specific, we prove the following:

Theorem B.1. *The MAX-INDEPENDENT-SET (MIS) problem with m vertices can be reduced to the Product-Distance model in $O(\text{poly}(m))$ time. Therefore, for every $\epsilon > 0$, no polynomial time algorithm can approximate the Product-Distance model within a factor of $m^{1-\epsilon}$ unless $P = NP$.*

Proof sketch. We defer the detailed proof of Theorem B.1 to the full version of the paper due to space limit, and only provide a proof sketch. Instead of directly reduce the MIS problem to the Product-Distance model, we introduce a new Maximum Sum of Exponential Degree (MSED) problem on an unweighted graph, which serves as a bridge between the MIS problem and the Product-Distance model. The problem is defined as follows:

Definition B.2 (Maximum Sum of Exponential Degree Problem). *Given an unweighted and undirected graph $G = (V, E)$, the Maximum Sum of Exponential Degree (MSED) problem aims at finding $\max_{U \subseteq V} \sum_{i \in U} 0.5^{\deg_U(i)}$, where $\deg_U(i)$ represents the number of vertices $j \in U$ such that edge $(i, j) \in E$.*

We show that the optimal solution for the MSED problem is always an independent set so the hardness of solving the MSED problem equals to the hardness of solving the MIS problem. To see this, consider any set S that is not an independent set, i.e. there exists $u, v \in S$ such that $(u, v) \in E$ and $\deg_S(u) \geq \deg_S(v)$. Then $\sum_{i \in S \setminus \{u\}} 0.5^{\deg_S(i)} - \sum_{i \in S} 0.5^{\deg_S(i)} = -0.5^{\deg_S(u)} + \sum_{u' \in S, (u, u') \in E} 0.5^{\deg_S(u')} \geq -0.5^{\deg_S(u)} + 0.5^{\deg_S(v)} \geq 0$. Thus one can remove u from S to weakly increase the objective.

Next, given a graph $G = (V, E)$, we further reduce the MSED problem to the Product-Distance model by constructing a Product-Distance instance with $n = |V|$ unweighted advertisers and $m = |V|$ slots, such that each vertex in G maps to a slot in $[m]$. For each $j_1, j_2 \in [m]$, we define $d(j_1, j_2) = 0.5$ if edge $(j_1, j_2) \in E$, and otherwise we define $d(j_1, j_2) = 1$. d is clearly a metric and we show that a good solution that approximates this Product-Distance instance can be converted back to a good solution for the MSED problem. Then, the hardness of the MSED problem implies the hardness result for the Product-Distance model.

C Omitted Proofs in Section 2

C.1 Proof of Lemma 2.3

PROOF OF LEMMA 2.3. Let the Nearest-Neighbor instance be given by advertisers $[n]$, slots $[m]$, nonnegative valuations $v_{i,j}$, and a normalized metric $d : [m] \times [m] \rightarrow [0, 1]$ (page diameter 1).

We build an instance of the Grouped Pseudo-Disk Selection Problem on the same metric space (X, d) with $X = [m]$. For each advertiser $i \in [n]$ (this becomes group T_i) and each pair (j, j') with $j \in [m]$ and $j' \in [m] \setminus \{j\}$, we create a pseudo-disk $D_{i,j,j'} = (c_j, r = d(j, j')/2, w = v_{i,j} \cdot d(j, j'))$, where $c_j \in X$ represents the point of slot j in the metric space. Then, we have $|T_i| = m(m-1) \leq m^2$ for all i , and there are $K = n$ groups of pseudo-disks.

Proof of the first statement. Let T^* be the optimal selection that satisfies Constraint (I). We aim to show

$$\begin{aligned} \text{SW}(M^*) &:= \sum_{(i,j) \in M^*} v_{i,j} \cdot \min_{j' \in S(M^*) \setminus \{j\}} d(j, j') \\ &\leq \sum_{(i,j,j') : D_{i,j,j'} \in T^*} v_{i,j} \cdot d(j, j'). \end{aligned} \quad (10)$$

To show (10), we first define

$$T' := \left\{ D_{i,j,j'} : (i, j) \in M^* \wedge j' = \arg \min_{j' \in S(M^*) \setminus \{j\}} d(j, j') \right\}.$$

to be a selection of pseudo disks in the Grouped Pseudo-Disk Selection problem. Note that T' is a feasible selection, because if vertex $x \in X$ is covered by two disks $D_{i_1,j_1,j'_1}, D_{i_2,j_2,j'_2} \in T'$, by the definition of pseudo disks, we have

$$\begin{aligned} d(j_1, j_2) &\leq d(j_1, v) + d(j_2, v) \\ &< 0.5 \cdot (d(j_1, j'_1) + d(j_2, j'_2)) \\ &\leq 0.5 \cdot (d(j_1, j_2) + d(j_2, j_1)) = d(j_1, j_2), \end{aligned}$$

where the first line follows from triangle inequality, the second line follows from the definition of $D_{i_1,j_1,j'_1}, D_{i_2,j_2,j'_2}$, and the last line uses the fact that j'_1 is the vertex nearest to j_1 in $S(M^*)$. Since $j_2 \in S(M^*)$, there must be $d(j_1, j'_1) \leq d(j_1, j_2)$. Similarly, we have $d(j_2, j'_2) \leq d(j_2, j_1)$. Then, the assumption that v is covered by two different disks leads to a contradiction, and therefore T' is feasible. By arguing that $\text{Val}(T') \leq \text{Val}(T^*)$,

$$\begin{aligned} \sum_{(i,j,j') : D_{i,j,j'} \in T^*} v_{i,j} \cdot d(j, j') &\geq \sum_{(i,j,j') : D_{i,j,j'} \in T'} v_{i,j} \cdot d(j, j') \\ &= \sum_{(i,j) \in M^*} v_{i,j} \cdot \min_{j' \in S(M^*) \setminus \{j\}} d(j, j'), \end{aligned}$$

where the equality follows from the definition of T' .

Proof of the second statement. To prove the second statement, we aim to show for any feasible selection T that satisfies Constraint (II), there exists a feasible matching M such that

$$\text{SW}(M) \geq 0.5 \cdot \sum_{(i,j,j_1) : D_{i,j,j_1} \in T} v_{i,j} \cdot d(j, j_1). \quad (11)$$

To prove (11), we define

$$M := \{(i, j) : \text{there exists } j' \text{ such that } D_{i,j,j'} \in T^*\}.$$

Note that M is a feasible matching, as if an advertiser i is matched to two slots j_1, j_2 , it is in contrast to the constraint that $|T \cup T_i| = 1$ for the Grouped Pseudo-Disk Selection Problem. On the other hand, if a slot j is matched to two advertisers i_1 and i_2 , c_j must be covered by two different disks in T with v serving as the center, which is in contrast to the assumption that c_j can be covered by at most once. Therefore, M is feasible.

For $(i, j) \in M$, let j_1 be the point that satisfies $D_{i,j,j_1} \in T$. Such j_1 must be unique, as the center c_j can be covered by at most once. Let $j_2 = \arg \min_{j' \in S(M) \setminus \{j\}} d(j, j')$. Note that there must be $0.5 \cdot d(j, j_1) \leq d(j, j_2)$: since j_2 is a selected slot in M , there must be a corresponding disk in T that uses c_{j_2} as the center. If $0.5 \cdot d(j, j_1) > d(j, j_2)$ is also satisfied, it implies that c_{j_2} is also covered by the disk $D_{i,j,j_1} \in T$, which is in contrast to Constraint (II). Then, $0.5 \cdot d(j, j_1) \leq d(j, j_2)$ must be satisfied, which implies

$$\begin{aligned} 0.5 \cdot \sum_{(i,j,j_1): D_{i,j,j_1} \in T} v_{i,j} \cdot d(j, j_1) &\leq \sum_{(i,j) \in M} v_{i,j} \cdot \min_{j' \in S(M) \setminus \{j\}} d(j, j') \\ &=: \text{SW}(M). \end{aligned} \quad \square$$

C.2 Proof of Theorem 2.6

PROOF OF THEOREM 2.6. By Myerson's Lemma [26], it's sufficient to show that the algorithm for Theorem 2.1 is *monotone*, that is, given a Nearest-Neighbor instance, if we increase the values $\{v_{i,j}\}$ to $\{v'_{i,j}\}$ so that $v_{i,j} \leq v'_{i,j}$, the (expected) welfare from the matching with valuations $\{v_{i,j}\}$ is at most the (expected) welfare from the matching with valuations $\{v'_{i,j}\}$.

We first show that after reduction, the Grouped Pseudo-Disk Selection problem is monotone. This follows from two observations: firstly, the objective of (LP_{RELAX}) is monotone, because when the optimal solution of (LP_{RELAX}) is fixed, increasing the weight w_i of a disk does not change the feasibility of the LP solution, but the objective of the LP increases. Secondly, by (3), the output $\tilde{T}$ of Algorithm 1 guarantees that $\mathbb{E}[\text{Val}(\tilde{T})]$ exactly equals to $\frac{1}{9}$ times the objective of (LP_{RELAX}). Therefore, the Grouped Pseudo-Disk Selection problem is monotone.

Note that when reducing a Nearest-Neighbor instance to a Grouped Pseudo-Disk Selection instance, the monotonicity is preserved, i.e., increasing the value of $v_{i,j}$ leads to a larger weight w_i for the Grouped Pseudo-Disk Selection instance. Then, to prove Theorem 2.6, it only remains to show that the monotonicity is preserved when converting the solution $\tilde{T}$ back to a feasible matching. To achieve this, given $\tilde{T}$, we perform the modified conversion: for every $D_{i,j,j'} \in \tilde{T}$,

- Add (i, j) into $\tilde{M}$.
- Add an empty slot $\hat{j}$ that satisfies $d(j, \hat{j}) = d(j, j')/2$ and a virtual advertiser $\hat{i}$ that has 0 value for every slot into the instance, and add $(\hat{i}, \hat{j})$ into $\tilde{M}$.

To verify the feasibility of $\tilde{M}$, the feasibility of $\{(i, j) : D_{i,j,j'} \in \tilde{T}\}$ is guaranteed by Lemma 2.3, and the feasibility of $\{(\hat{i}, \hat{j})\}$ follows from the fact that both $\hat{i}$ and $\hat{j}$ are newly added to the instance. Furthermore, note that after adding $(\hat{i}, \hat{j})$ to the matching, each $(i, j) : D_{i,j,j'}$ contributes exactly $v_{i,j} \cdot d(j, j')/2$ to $\text{SW}(\tilde{M})$, which is 0.5 times the weight of $D_{i,j,j'}$, and each $(\hat{i}, \hat{j})$ contributes 0 to

$\text{SW}(\tilde{M})$. Then, we have $\text{SW}(\tilde{M}) = 0.5 \cdot \text{Val}(\tilde{T})$, and therefore the monotonicity is preserved. $\square$

D Omitted Proofs in Section 3

D.1 Proof of Claim 3.2

PROOF. The symmetricity of function f follows from the fact that f is applied to the sorted vector $Z^\downarrow$. Therefore, the order of values in Z does not effect the value of function f . For the monotonicity, consider two n -dimensional vectors $(z_1, \dots, z_m)$ and $(y_1, \dots, y_m)$ that satisfies $z_i \geq y_i$. Then, there must be $z_i^\downarrow \geq y_i^\downarrow$: there are at least i values in vector y being at least $y_i^\downarrow$, and therefore there are at least i values in vector z being at least $y_i^\downarrow$, which implies $z_i^\downarrow \geq y_i^\downarrow$. Then, we have

$$f(Z; W^\downarrow) - f(Y; W^\downarrow) = \sum_{i \in [m]} (z_i - y_i) \cdot w_i^\downarrow \geq 0,$$

and therefore the monotonicity holds.

It remains to show that function f is a norm. We have the following:

- Positivity: The non-negativity of f is guaranteed by the non-negativity of vector Z and the vector $W^\downarrow$. We further assume that $w_1^\downarrow > 0$, as otherwise it implies all advertisers have value 0, making the problem trivial. Then, $f(Z, W^\downarrow) \geq z_1^\downarrow \cdot w_1^\downarrow$, and therefore $f(Z, W^\downarrow) = 0$ implies vector Z only contains value 0.
- Absolute Homogeneity: For any $\alpha \geq 0$, we have

$$f(\alpha \cdot Z; W^\downarrow) = \sum_{i=1}^m (\alpha z_i^\downarrow) \cdot w_i^\downarrow = \alpha \cdot f(Z, W^\downarrow).$$

- Triangle Inequality: Let $(z_1, \dots, z_m)$ and $(y_1, \dots, y_m)$ be two n -dimensional vector that are not necessarily ordered. As function f is symmetric, we assume without loss of generality that $z_1 + y_1 \geq z_2 + y_2 \geq \dots \geq z_m + y_m$. We assume without loss of generality that $w_{m+1}^\downarrow := 0$, as $w_{m+1}^\downarrow$ is not needed in function f . Since $w_i^\downarrow \geq w_{i+1}^\downarrow$ for every $i \in [n]$, we have

$$\begin{aligned} f(Y + Z; W^\downarrow) &= \sum_{i=1}^m (y_i + z_i) \cdot w_i^\downarrow \\ &= \sum_{i=1}^m \left((w_i^\downarrow - w_{i+1}^\downarrow) \cdot \sum_{k=1}^i (y_k + z_k) \right) \\ &= \sum_{i=1}^m \left((w_i^\downarrow - w_{i+1}^\downarrow) \cdot \sum_{k=1}^i y_k \right) + \sum_{i=1}^m \left((w_i^\downarrow - w_{i+1}^\downarrow) \cdot \sum_{k=1}^i z_k \right) \\ &= \sum_{i=1}^m y_i \cdot w_i^\downarrow + \sum_{i=1}^m z_i \cdot w_i^\downarrow \\ &\leq f(Y; W^\downarrow) + f(Z; W^\downarrow), \end{aligned}$$

where the inequality holds as it becomes an equality only when vector Y and Z are both sorted.

Therefore, function f is a norm. $\square$

D.2 Constant Approximation for Stochastic Advertiser Values

In this subsection, we show that the approximation ratio for the Nearest-Neighbor model is a constant when value $v_{i,j}$ can be factorized as $v_{i,j} = w_i \cdot u_j$, while each w_i is drawn from distribution $\mathcal{D}_i$. To be specific, we prove the following:

Theorem D.1. *For Nearest-Neighbor model, assuming each advertiser's valuation factorizes as $w_i \cdot u_j$, where w_i is a private value of advertiser i independently drawn from a publicly known distribution $\mathcal{D}_i$, and u_j is a public weight associated with slot j . Then, there exists an allocation algorithm running in $\tilde{O}(n + m)$ time that achieves a 504-approximation. Furthermore, the algorithm can be converted to a truthful mechanism.*

Due to space limit, we provide the main algorithm and a proof sketch for Theorem D.1, and defer the detailed proofs to the full version of the paper.

We first assume without loss of generality that $m \leq n$, as we can arbitrarily add advertisers with zero value into the instance. The main idea of the algorithm is that the slots can be pre-selected by resolving the instance with slot values $\{u_j\}$ and advertiser values $\{\gamma^{(k)}\}$, where we define

$$\gamma^{(k)} := \mathbb{E}_{X_i \sim \mathcal{D}_i} [X_k^\downarrow] \quad (12)$$

to be the expectation of the k -th largest value $X_k^\downarrow$ of samples $X_1, \dots, X_n$. Formally, we present our algorithm in Algorithm 3.

Algorithm 3: CONSTANT APPROXIMATION FOR FACTORIZED NEAREST-NEIGHBOR MODEL

Input: Number of slots m and advertisers n , distributions $\mathcal{D}_1, \dots, \mathcal{D}_n$, distance function d
 /* Slot Selection Phase */
 Input slot weights $u_1, u_2, \dots, u_m$, such that WLOG $u_1 \geq u_2 \geq \dots \geq u_m$.
 For $k \in [n]$, calculate $\gamma^{(k)}$ defined in (12).
 Solve the Nearest-Neighbor instance via the algorithm from Theorem 2.1 with values $v_{i,j} = u_j \cdot \gamma^{(i)}$ and distance function $d(\cdot, \cdot)$. Let $\hat{M}$ be the returned matching.
 Let $\tilde{S} = S(\hat{M})$ be the pre-selected slots.
 /* Allocation Phase */
 Advertisers report values $w_1, w_2, \dots, w_n$, such that $w_i \sim \mathcal{D}_i$.
 For $i \in \min\{|\tilde{S}|, n\}$, assign the advertiser with i -th largest value to the slot in $\tilde{S}$ such that the value of $u_j \cdot \min_{j' \in \tilde{S} \setminus \{j\}} d(j, j')$ is the i -largest.
 Let $\tilde{M}$ be the corresponding matching.
Output: Matching $\tilde{M}$.

Proof sketch of Theorem D.1. We first represent the objective of $\tilde{M}$ given by Algorithm 3 as a norm. Given an instance of the factorized Nearest-Neighbor model where $w_1, w_2, \dots, w_n$ are values of advertisers such that $w_i \sim \mathcal{D}_i$, and $u_1 \geq u_2 \geq \dots \geq u_m \geq 0$ are weights of slots. For n -dimensional vector $W = (w_1, w_2, \dots, w_n) \in \mathbb{R}_{\geq 0}^n$ that only contains non-negative values and subset $S \subseteq [m]$,

we define function $g_S : \mathbb{R}_{\geq 0}^n \rightarrow \mathbb{R}_{\geq 0}$ as follows:

$$g_S(W) = \sum_{i=1}^m w_i^\downarrow \cdot (Z^\downarrow(S))_i,$$

where we recall function $Z(S)$ returns a vector, such that $(Z(S))_j = 1[j \in S] \cdot u_j \cdot \min_{j' \in S \setminus \{j\}} d(j, j')$. Then, the value of $g_S(W)$ represents the objective of the optimal matching, assuming vector W contains advertisers' values, and subset S is the selected subset of slots, i.e., let M^* be the optimal matching, we can simplify the value of $\text{SW}(M^*)$ as $\max_{S \subseteq [m]} g_S(W)$, and we further define

$$h(W) := \max_{S \subseteq [m]} g_S(W).$$

Then, the following claim suggests that both $g_S(W)$ and $h(W)$ are norm functions, where the proof is omitted due to space limit:

Claim D.2. *Function $g_S(W)$ with any $S \subseteq [m]$ and $h(W)$ are both monotone symmetric norms.*

Now, we are ready to show the approximation ratio for Algorithm 3. Our proof relies on the following lemma:

Lemma D.3 (Theorem 5.1 in [21]). *Let vector $W \in \mathbb{R}_{\geq 0}^n$ follow a product distribution on $\mathbb{R}_{\geq 0}^n$, and $f : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ be a monotone symmetric norm. Then $f(\mathbb{E}[W^\downarrow]) \leq \mathbb{E}[f(W)] \leq 28 \cdot f(\mathbb{E}[W^\downarrow])$.*

We apply Lemma D.3 with $W = (w_1, \dots, w_n)$ being the vector of advertisers' values. Note that vector W follows a product distribution, such that $w_i \sim \mathcal{D}_i$, and vector $\mathbb{E}[W^\downarrow]$ is exactly $(\gamma^{(1)}, \gamma^{(2)}, \dots, \gamma^{(n)})$. For simplicity of the notation, we define vector $\Gamma = \mathbb{E}[W^\downarrow] = (\gamma^{(1)}, \dots, \gamma^{(n)})$.

Recall that when advertiser's value vector W is fixed, the objective $\text{SW}(M^*)$ is defined as $h(W)$. By taking the expectation over randomness of W and applying Lemma D.3, we have

$$\mathbb{E}[\text{SW}(M^*)] = \mathbb{E}[h(W)] \leq 28 \cdot h(\Gamma),$$

where the inequality uses the fact that function h is a monotone symmetric norm. A key observation is that $h(\Gamma)$ is exactly the objective of the optimal matching of the instance we constructed in Line 3 of Algorithm 3. Let function $\widehat{\text{SW}}(M)$ be the objective of matching M for this instance, and $\hat{M}^*$ be the optimal matching of this instance. Then, we have

$$h(\Gamma) = \widehat{\text{SW}}(\hat{M}^*) \leq 18 \cdot \widehat{\text{SW}}(\hat{M}) \leq 18 \cdot g_{\tilde{S}}(\Gamma),$$

where the first inequality follows from the fact that the algorithm in Theorem 2.1 gives a 18-approximation, and the second inequality follows from the fact that $\tilde{S} = S(\hat{M})$, and when $\tilde{S}$ is fixed, the optimal matching is nothing but matching values in Γ and $Z(\tilde{S})$ in descending order.

Finally, since function $g_{\tilde{S}}$ is a monotone symmetric norm,

$$g_{\tilde{S}}(\Gamma) \leq \mathbb{E}[g_{\tilde{S}}(W)].$$

Merging the above three inequalities together gives

$$\mathbb{E}[\text{SW}(M^*)] \leq 28 \cdot h(\Gamma) \leq 28 \cdot 18 g_{\tilde{S}}(\Gamma) \leq 504 \cdot \mathbb{E}[g_{\tilde{S}}(W)],$$

which proves the approximation ratio of Algorithm 3, as $\mathbb{E}[g_{\tilde{S}}(W)]$ exactly represents the performance of Algorithm 3.

It remains to prove the truthfulness and the running time of Algorithm 3. We defer the proofs to the full version of the paper.